



PSMC PDK usage guide:
An introduction on the usage of PSMC
process design kits (PDK)

Product Version: V1.0

2019.01

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1 Introduction

The major purpose of this document is to introduce the basic usage of a PDA's PDK for those users who are completely new to PDA PDK or never use PDA's PDKs before as a reference. To ease the overall introduction, we use a simple design as an example to go through the whole design flow: starting from the schematic capture and ending at the physical verification and post-layout simulation. Moreover, we divide the whole flow into several phases to match general logic or Mix-signal design flows and they are "Schematic capture", "Analog simulation", "Physical Layout", "Custom Route", "Physical verification" and "Extraction and Parasitic re-simulation".

Each phase may contain various kinds of flows corresponding to different tools. Users can choose those flows according to their needs. For example, in the Analog simulation phase, two kinds of simulators can be chose to carry out the simulation. You only need to choose one simulator for your simulation.

This document doesn't focus on the tool usage. For the detail usage of the related tools, please contact the tool vendors.

2 Schematic capture

After you have finished the installation of a PDA's PDK, we will start to create a new design based on the installed PDK.

2.1 Environment setup

Before we start to create a new design, some environment setups should be done.

First, we have to set the environment variable of "CDS_Netlisting_Mode" to "Analog". This can be archived by the following UNIX command:

```
setenv CDS_Netlisting_Mode "Analog"
```

Then, try to find a directory to create your project by:

```
%mkdir ~/my_project
```

```
%cd ~/my_project
```

All the further designs created by users should be performed in this directory.

The next steps are to copy some necessary files from the PDK installed directory to your project directory. These include copying the 'display' file and linking the 'Model' and others to your project directory.

```
%cp <pdk_install_directory>/display.drf .
```

```
%ln -s <pdk_install_directory>/Model .
```

```
%cp -f <pdk_install_directory>/assura_tech.lib . (For Assura tool)
```

```
%cp -rf <pdk_install_directory>/Assura . (For Assura tool)
```

```
%cp -rf <pdk_install_directory>/Calibre . (For Calibre tool)
```

The last step for environment setup is to create a "cds.lib" file in your project directory. Users can use a text editor to add the following line into the "cds.lib" file located in your project directory:

```
INCLUDE <pdk_install_directory>/cds.lib
```

Note:

1. The installation procedures of a PDA's PDK can be found in the document of "PDA PDK reference manual " released along with the corresponding PDK.
2. The <pdk_install_directory> is referred to the path where the PDA's PDK was installed.

2.2 Creating a Library

After completing the environment setup, we can start to create a new library. This can be achieved by using the command “File->New->Library” from either the CIW (Command Interpreter Window) or the Library Manager and select the “Attach to an existing techfile” option. When the Library Manager asked for the name of the “Attach to Technology Library”, please select the name of PDK library that you installed for this design.

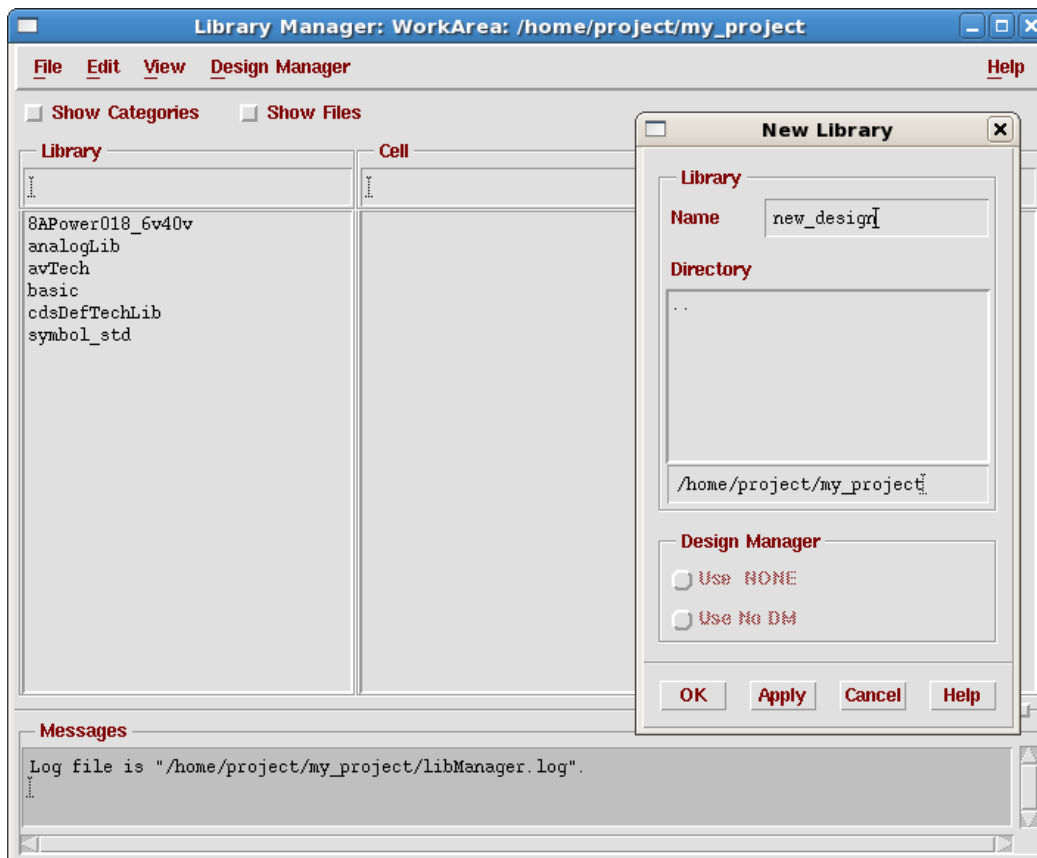


Fig. 1

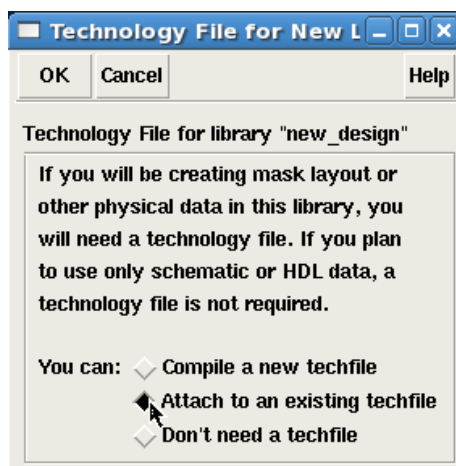


Fig. 2

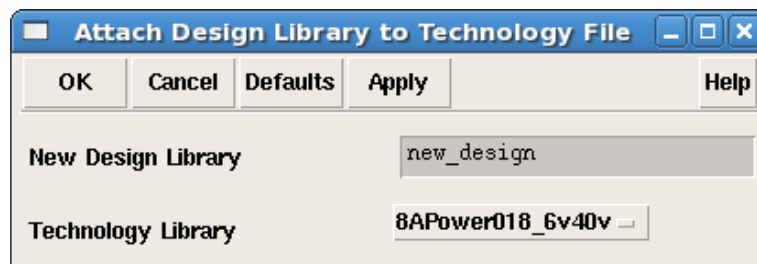


Fig. 3

2.3 Creating a design

When the library creation is completed, we can now start to create our design circuit under Cadence Composer environment. The basic steps to capture a new schematic including: open a new schematic design, select and place components, edit component properties and wire components...and so on.

From now on, we will use a one-stage invert as an example, which was designed based on PSMC PDK, to continue the following introductions in this document.

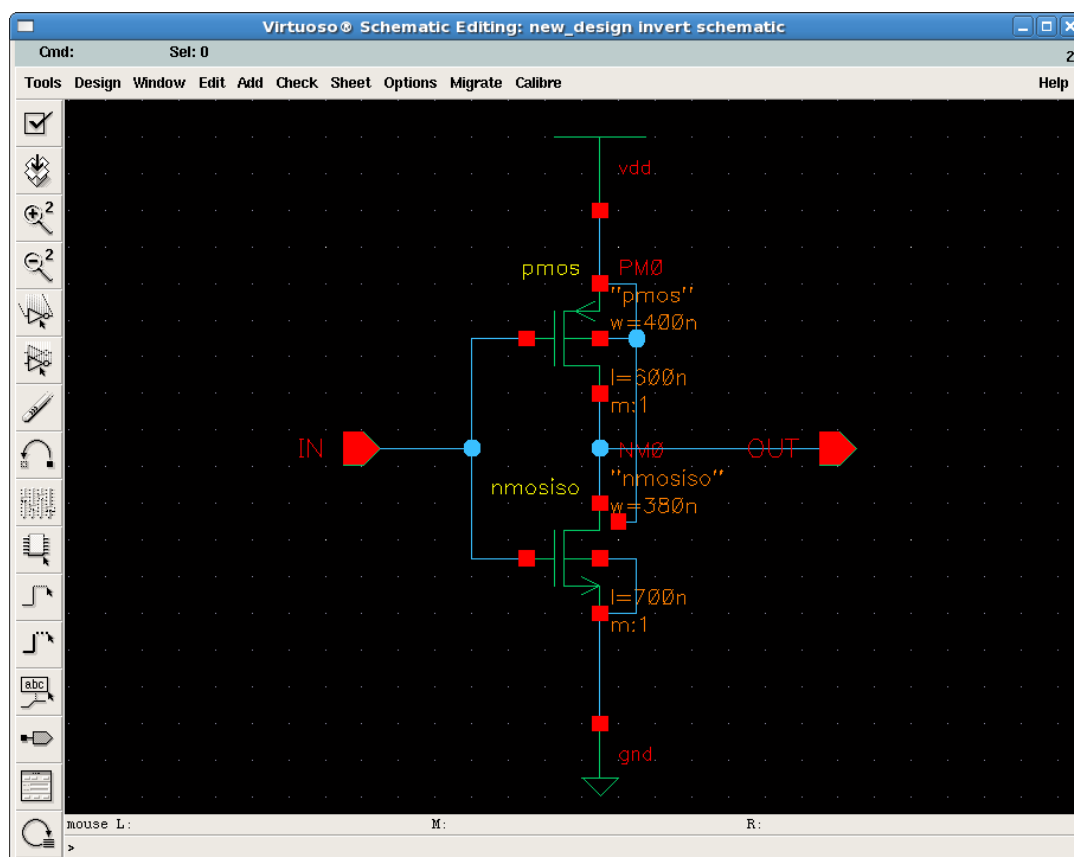


Fig. 4 schematic-capture of a inverter

2.4 Creating a symbol

After completing the creation of schematic-capture, we have to create its corresponding symbol for the PSMC PDK Usage Guide v1.0 2019.01

subsequence simulation steps.

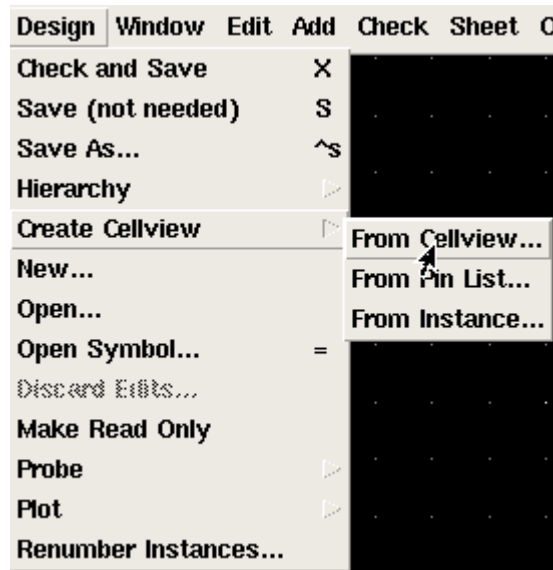


Fig.5

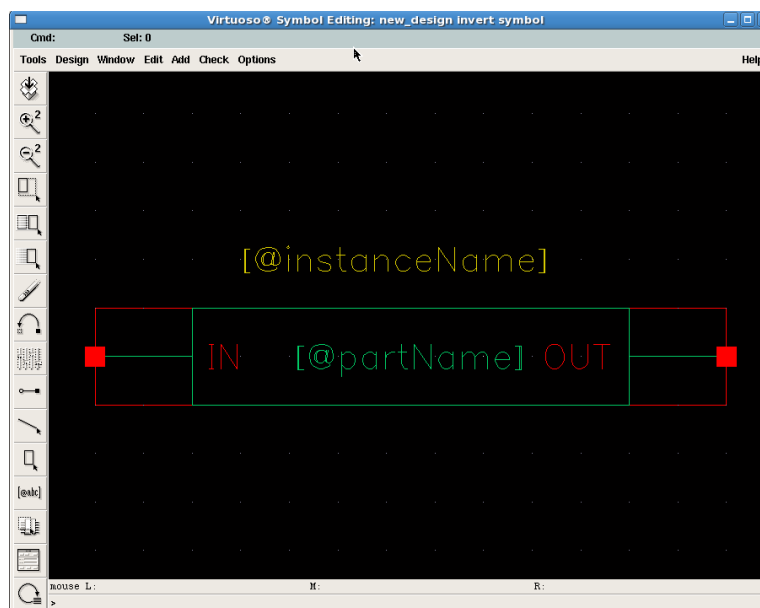


Fig.6 symbol for the inverter

2.5 Creating a test design

The final step before we start the simulation is to create a test fixture for our design. The creation of test fixture is similar to the creation of a design.

Furthermore, you also have to prepare the voltage source and determine the output loading. Generally, a test fixture will consist of the following components:

a design (the buffer in our case), DC voltage source, ground, vdd, voltage, source(vdc/vpluse), and output loading. The test fixture that we used for our design is namely 'test_invert' and can be seen in Fig. 7.

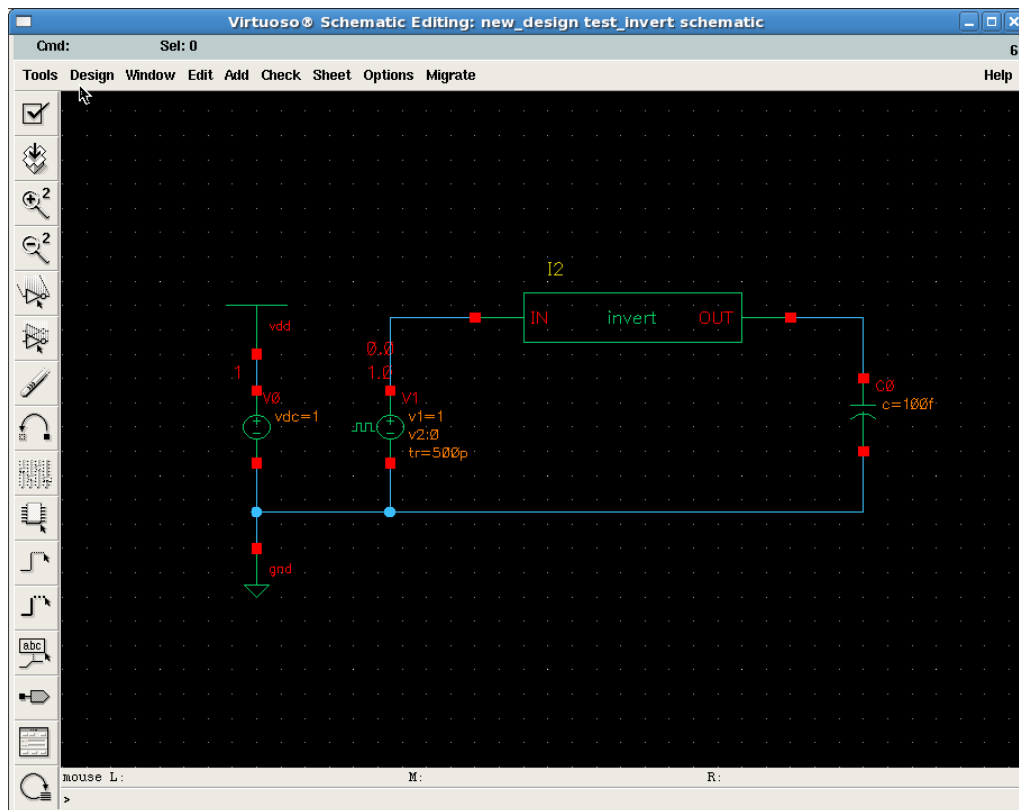


Fig. 7

3 Analog simulation

3.1 Performing pre-simulation

After the creation of our design is completed, we have to verify the electrical performance and the functionality of our design using a simulation tool. In this chapter, we will have introductions on two simulators: Spectre and Hspice.

3.2 Using hspice for simulation

In this section, we will start to simulate our design with spectre simulator under Analog Artist environment. The simulation step for hspice is as follows:

3.2.1 Open Analog Artist window

This can be archived by “Tools->Analog environment” in the menu banner of schematic view (Fig. 8).

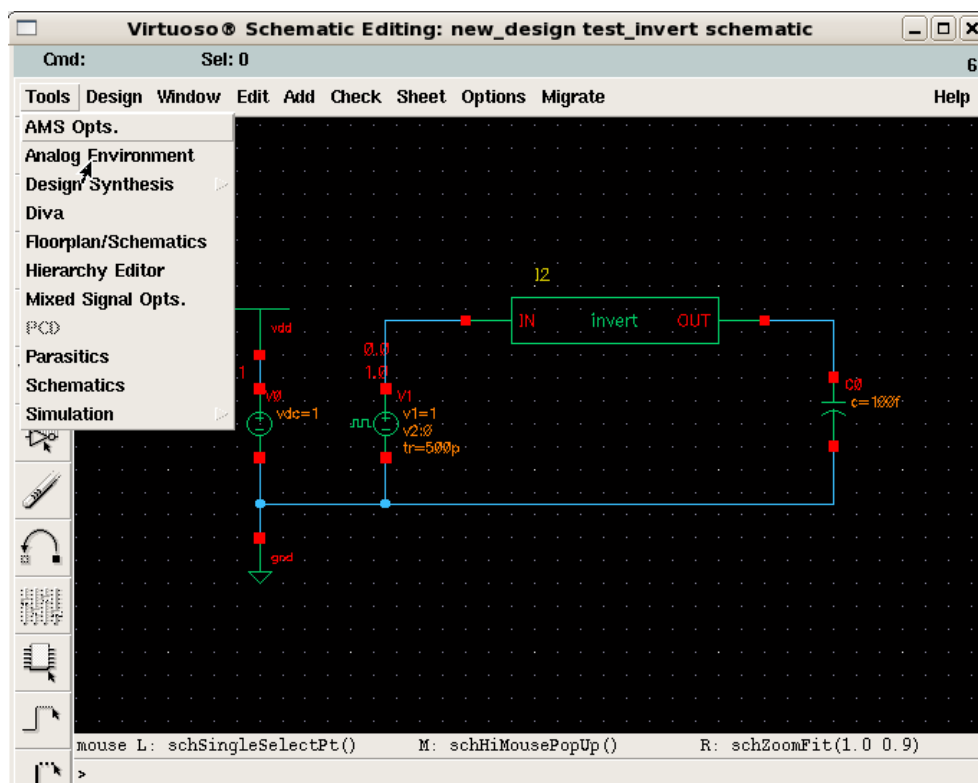


Fig.8

3.2.2 Choose hspice simulator and set hspice models

Open “Tools->Analog environment” in the menu banner of schematic view (Fig. 8). The default simulator is spectre, switch it to hspice by “Setup->Simulator/Directory/Host...”:

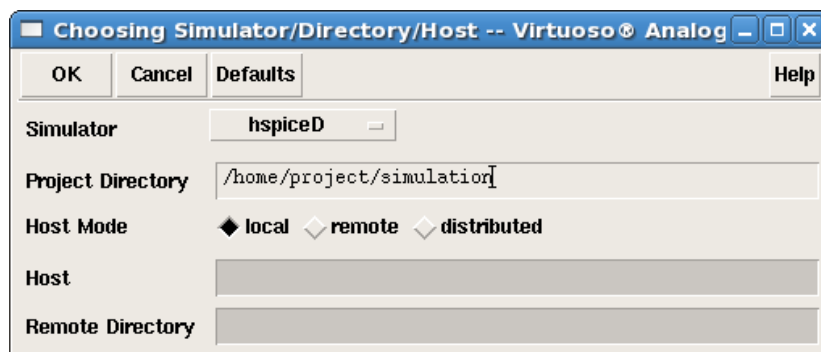


Fig.9

After the selection of simulator, the models should be set for hspice:

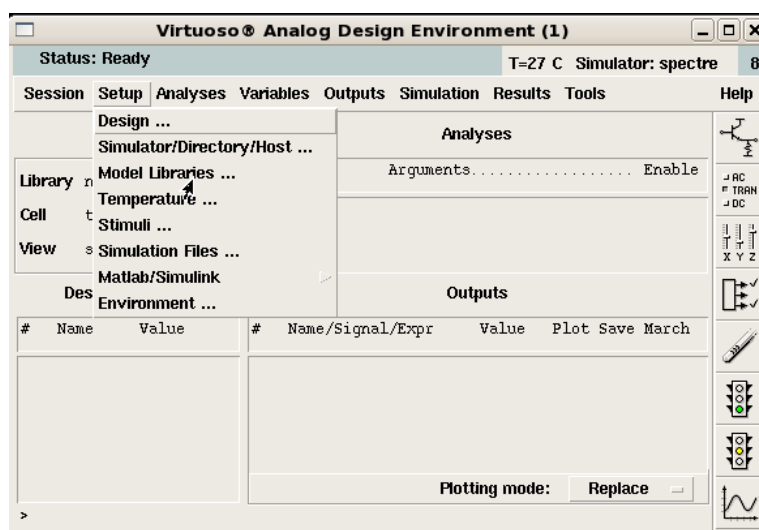


Fig.10

3.2.3 Select analysis type and fill in parameters for simulation

In the Analog Artist window, there are many analysis options that you can choose. Since we want to analysis the delay information (performance) of our design, we choose the transient analysis for our design. Some designers may want to see the OP point and they can also include the DC op point analysis.

After choosing Transient Analysis, we should set the simulation start/stop time and step:

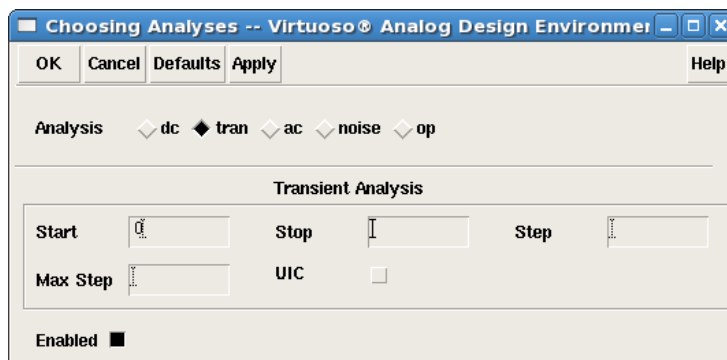


Fig.1

In addition to choosing the analysis type, we also need to select the nodes that we want to observe as simulation results from the schematic view. In our test fixture, we choose the “IN” and “OUT” nodes of our buffer circuit for the result browsing.

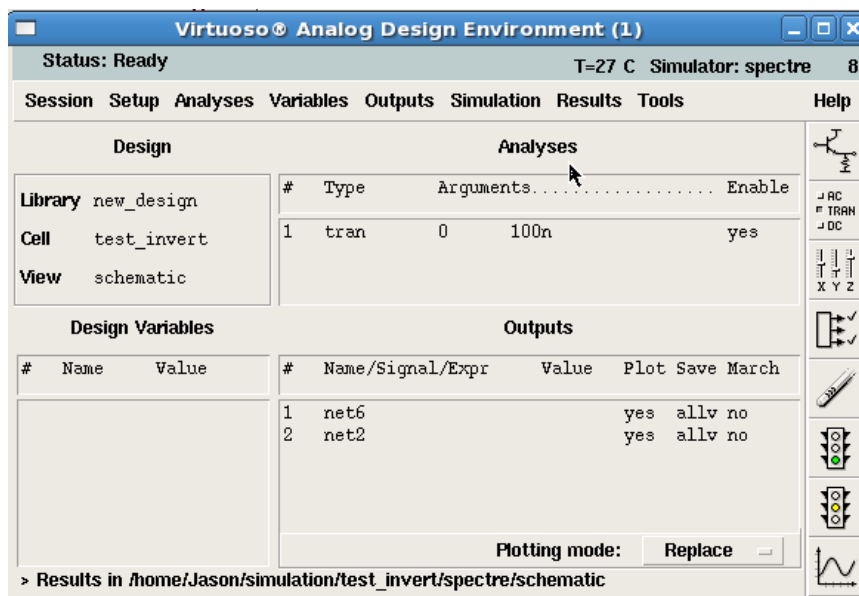


Fig.12

3.2.4 Run simulation and observe the simulation results

To start the simulation, you can click “simulation->netlist and run” from the Analog Artist menu banner. After the simulation is completed, the wave form window appears. You can see the simulation result that corresponding to the chose nodes. In our case, the waveform of nodes “IN” and “OUT” of our design circuit (invert) is shown.

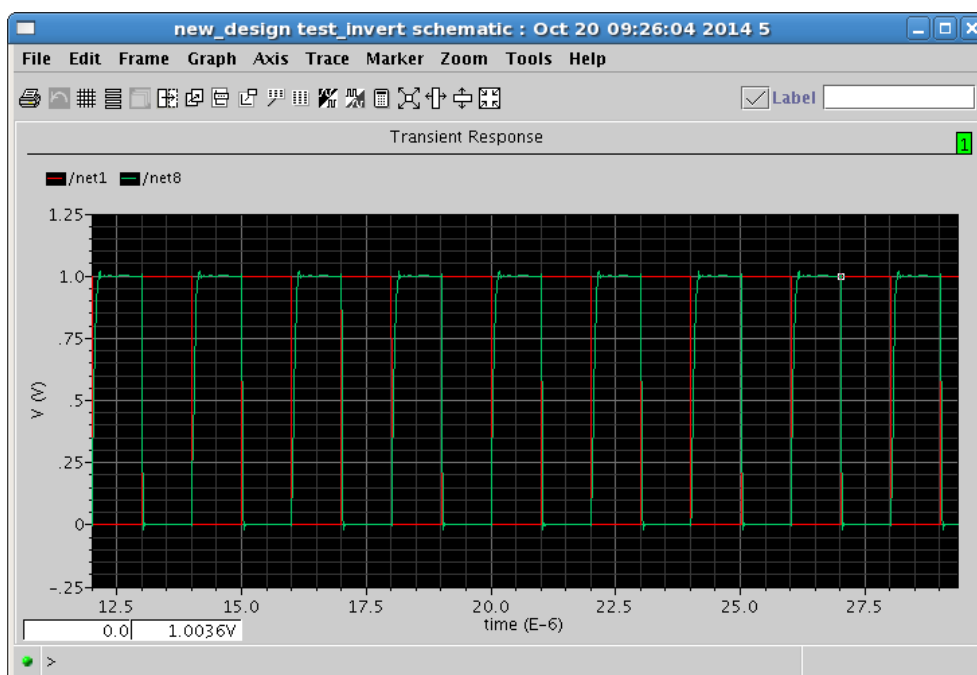


Fig.13

3.2.5 Annotating simulation results back to schematic

Sometime, you may want to see more detail simulation results such as operating points, DC voltages, and model parameters on the schematic of your design. You can back annotate the simulation data to the schematic by choose “Results->Annotate->...” from the menu banner of Artist simulation window. The available choices of annotated data are “DC node voltages”, “DC operating points”, “Transient node voltages”, “Transient operating points”, “Model parameters”, “Component parameters”...and so on.

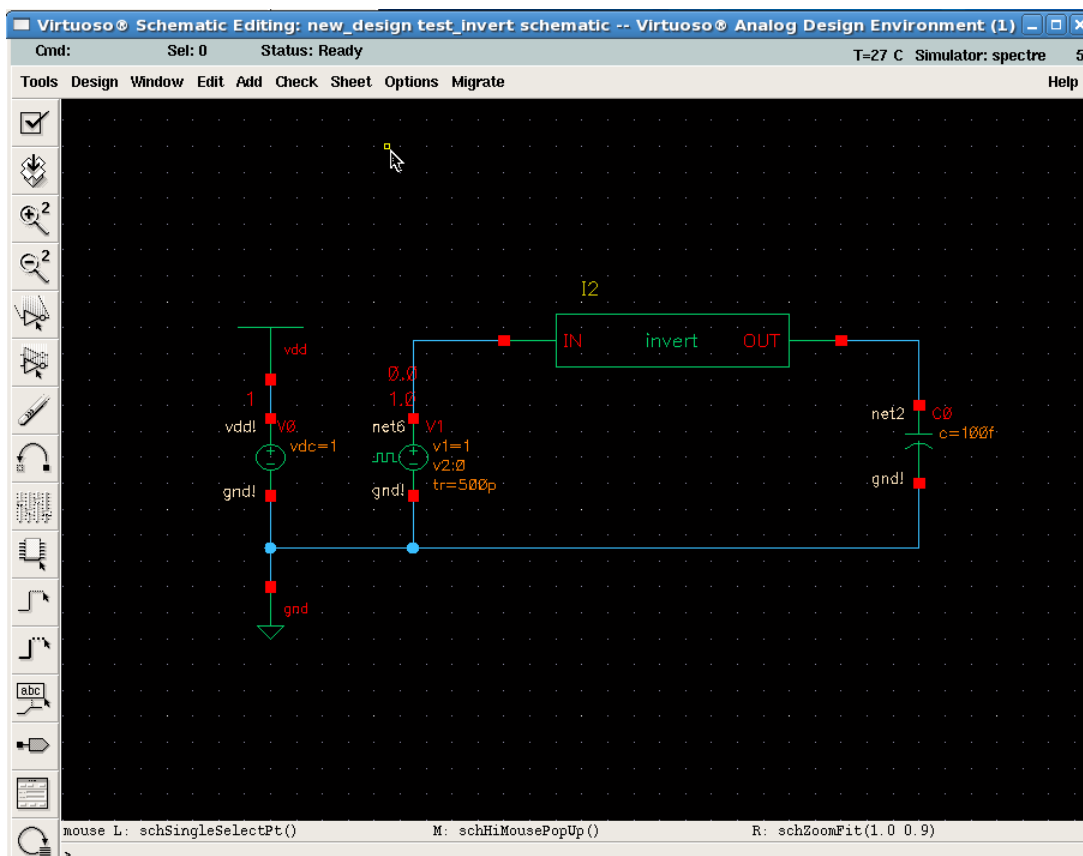


Fig.14

4 Physical Layout

After completed the pre-layout simulation and make sure the functionality and the circuit performance are all correct and in the design specifications, we can now start to create the corresponding layout for our design. There are many ways to create the layout including the tradition manual way and other automatic ways.

Generally, the tradition layout creation method is to draw the layer geometries based on schematic manually, which is usually labor and time-consuming. To take the advantage of using a PDK, we use another more efficient way to create the layout for our design. The layout creation procedures for our automatic way are partitioned into three parts: “Schematic-Driven-Layout”, “Components Placement”.

4.1 Schematic- Driven- Layout

Steps for Schematic-Driven-Layout method:

1. Open the schematic view of our design
2. From the schematic menu select “Tools -> Design Synthesis -> Layout XL”

After selecting this option, a small dialog box will first open to let users select the cell name and view name for the layout. Upon finished the selection of the cell name and view name, a Virtuoso XL layout window pop up for layout generation.

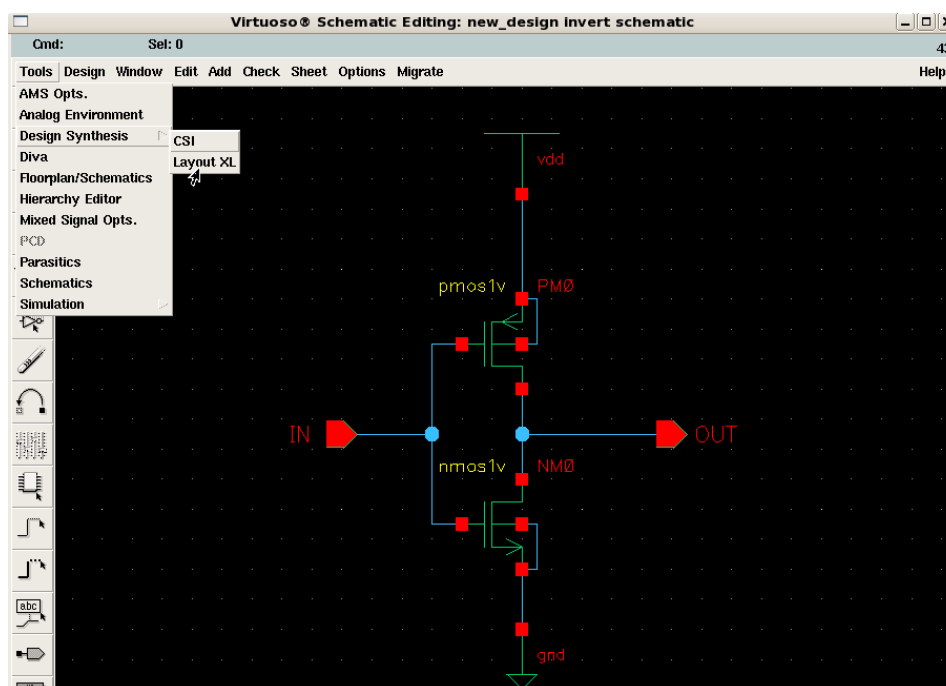


Fig.15

3. From the Virtuoso XL layout menu select “Design -> Gen from source ...”

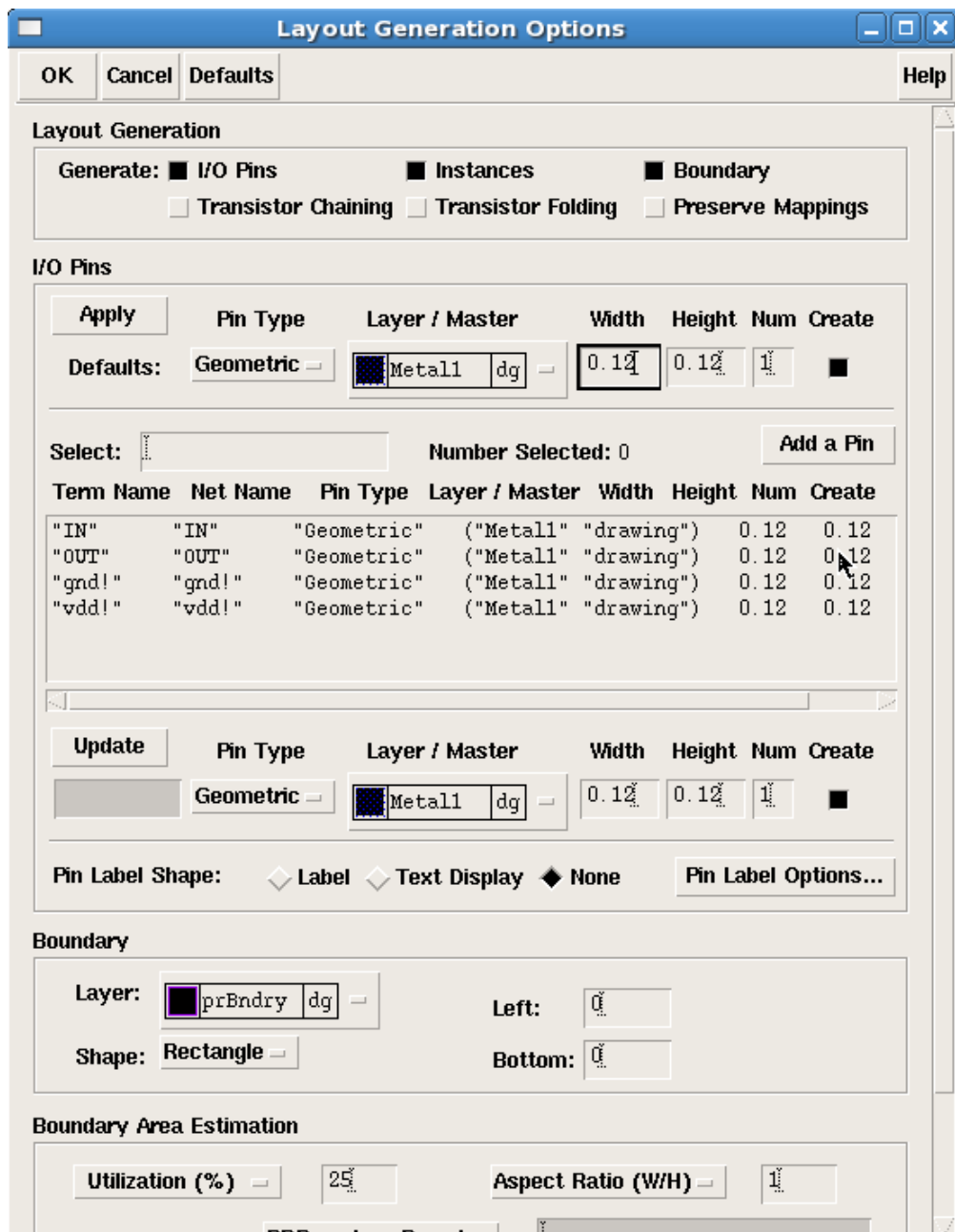


Fig.16 layout generation options window

After finished the selection of above information, some rectangles that represent the transistors and I/O pins will show up in the bottom of the layout window.

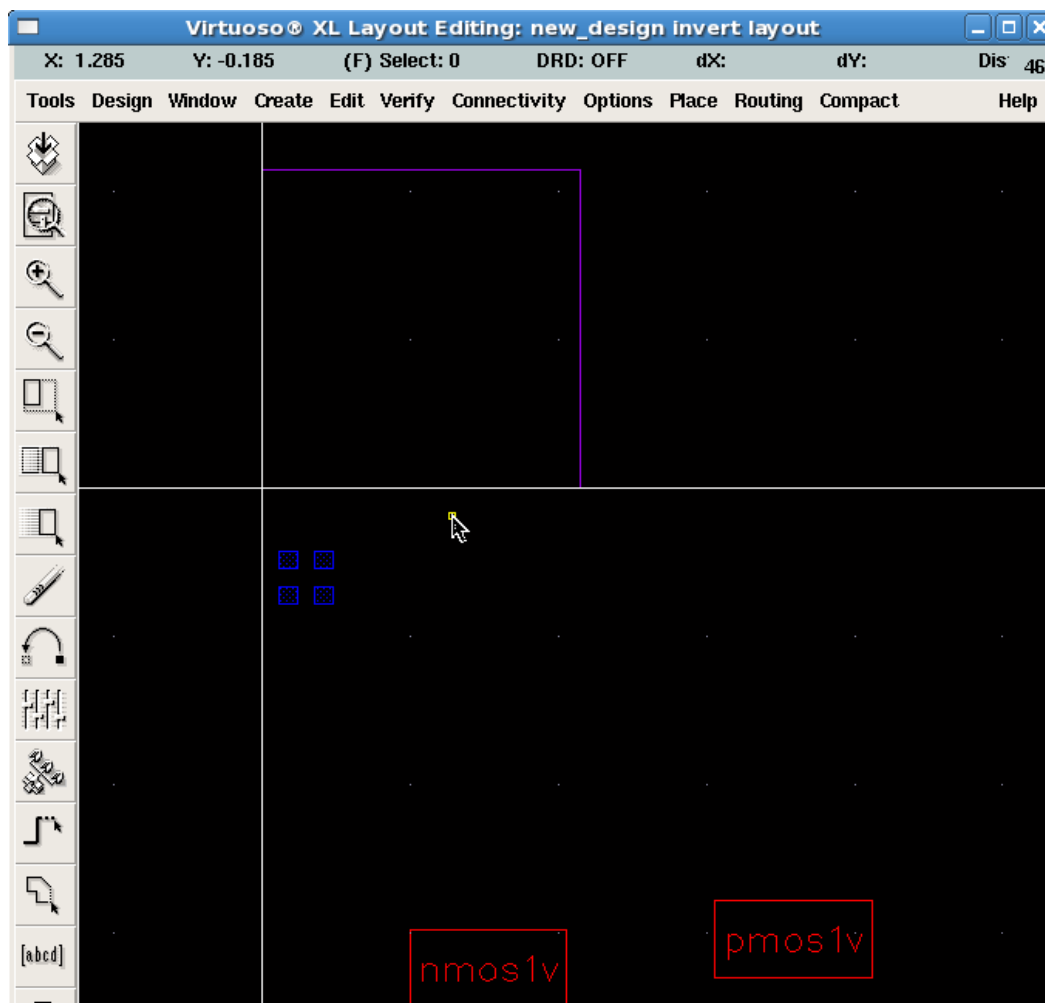


Fig.17

4.2 Components placement

The next step is to do the device placement. The only one thing that you need to do is to place all the components and I/O pins in the layout window into the design area (cell boundary). By selecting devices/I/O pins and dragging them to proper locations inside the design area, we can complete the component placement. During the device movement and placement, the lines represent the connections of select object to other objects will show up. This can help you to decide where to properly locate the selected object (Fig 18).

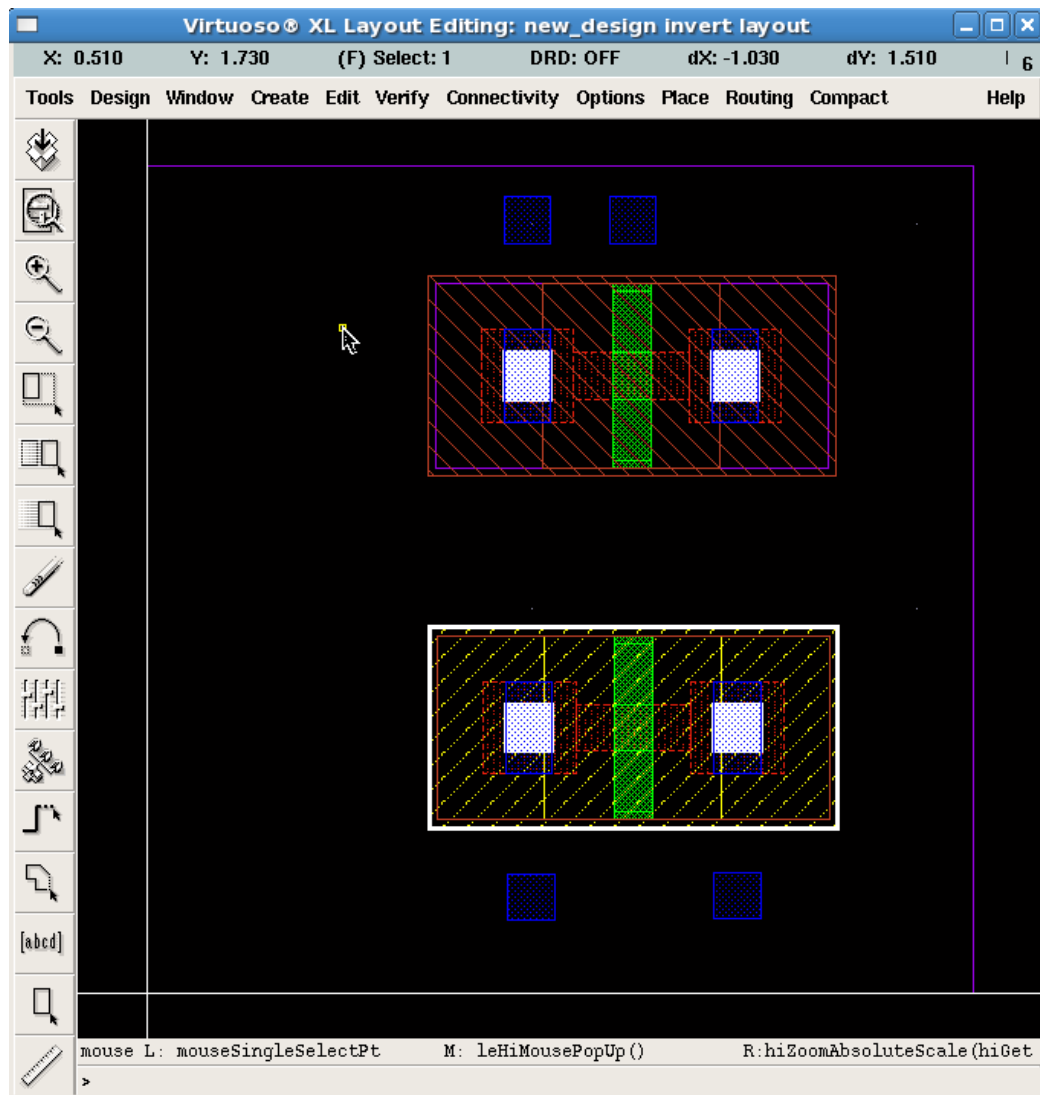


Fig.18

5 Auto-route and Manual route

When the component placement is completed, the next step is to perform the device routing. Since PDA's PDK also support the ICC rules for Cadence Virtuoso Custom Router (VCR), we will use the VCR to take full advantage of automated routing capability. After that, if we don't feel comfortable with the routing results, we can perform some manual routes to fix the routing results.

The steps for auto route are as follows:

1. Export the design to the router by Click "Route->Export to router"
2. Specify the ICC rules and select the router to "Virtuoso custom router" in the popup window to initiate the VCR (Fig. 19).
3. Click "Auto route->Detail route" to complete the auto routing (Fig. 20).

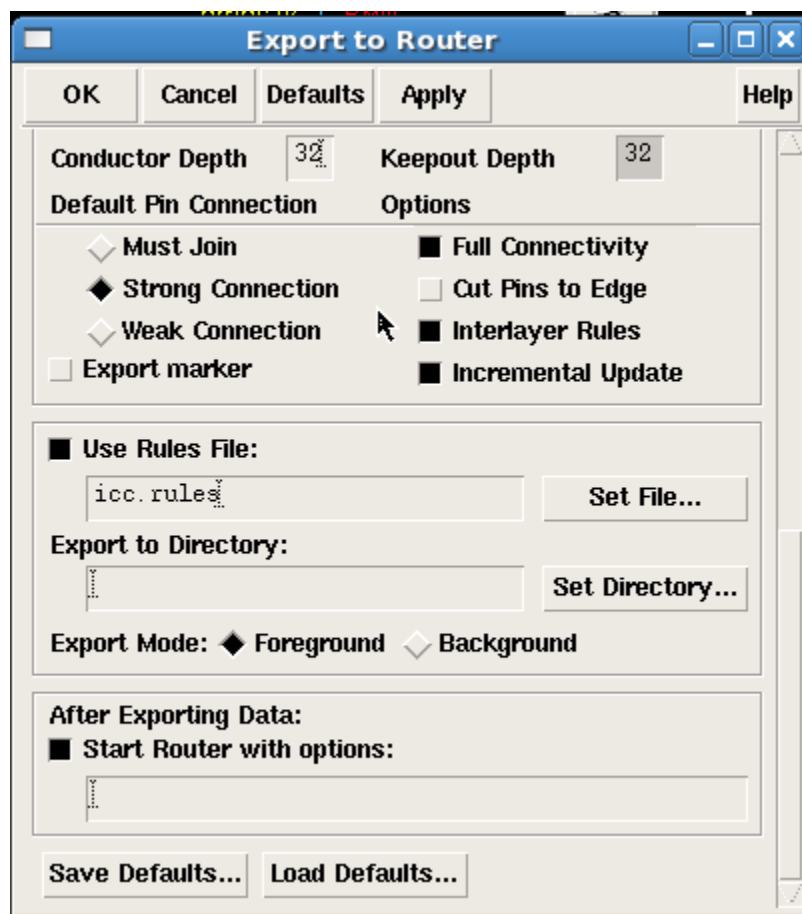


Fig.19

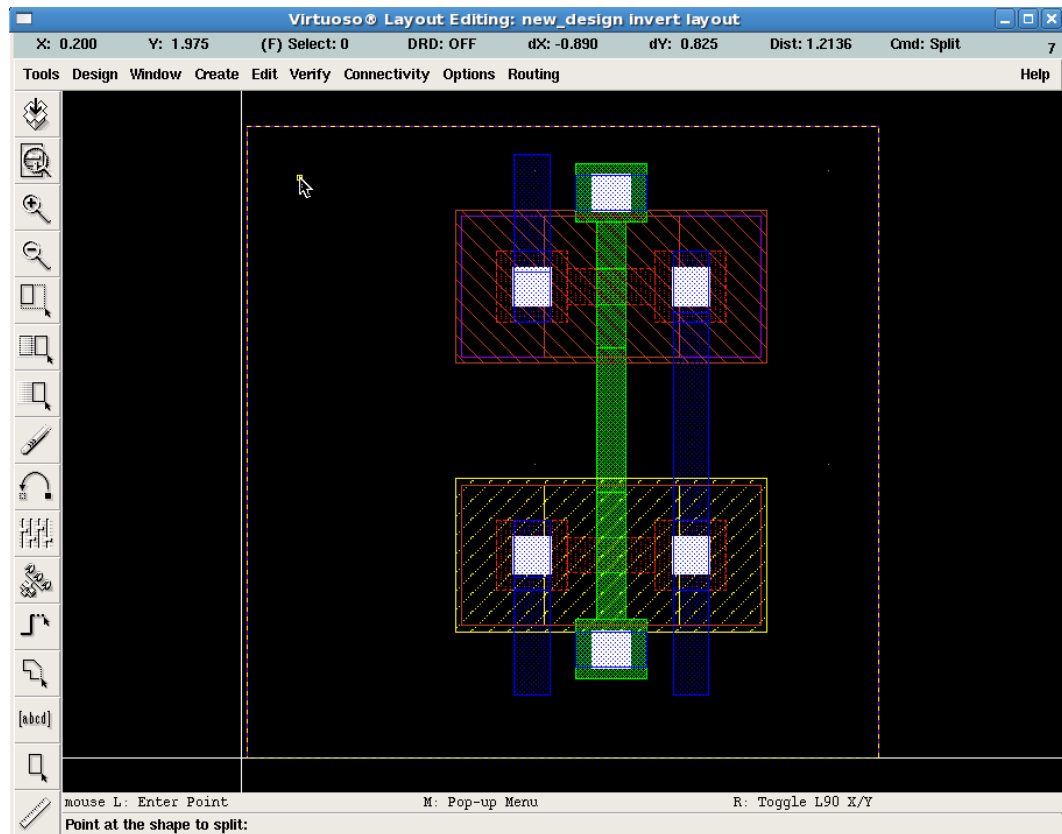


Fig.20

6 Physical verification

After the layout creation is completed, we have to start the physical verification to make sure this layout is DRC free and each device in the layout is completely match to its corresponding component in original schematic. After that, the parasitic extraction is also need to be performed for post-layout simulation to make sure our design still work well after taking the parasitic R & C effects into account. Generally, the physical verification procedures can be divided into three parts: the design rule check (DRC), layout V.S. schematic check (LVS) and parasitic extraction (RCX). Furthermore, based on the differences on running methods and supported tools, they can be classified into different flows. In order to satisfy most of users, PDA's PDK supports varied kinds of decks to be used for different kinds of flows. Currently, the supported tools are Assura and Calibre, and the supported flows are "Assura DFII flow", "Assura CDL flow", "Calibre CDL flow" and "Calibre interactive flow". In real design, users only need to choose one of these physical verification flows and have no need to go through all the flows.

6.1 Physical verification using Assura

In this section, we will introduce the "Assura DFII flow" and "Assura CDL flow" for Assura users.

6.1.1 Assura DFII flow

When choosing Assura as the physical verification tools, most of people will go through their physical verification with DFII flow because of its conveniency and user friendly. The physical verification procedures for Assura DFII flow are as follows:

6.1.1.1 Assura DFII DRC

1. Click "Assura->Run DRC..." in layout window to invoke Assura DRC graphic user interface.
2. Fill in the "Assura run directory" and select the "Technology" field to "assura_tech" (Fig. 21) in Assura DRC window. You can also set the Assura DRC switches and other parameters if needed.
3. Click "OK" to run the Assura DRC and see the result. If the layout isn't DRC free, you have to manually re-edit the layout and re-run the DRC check to make it DRC free.

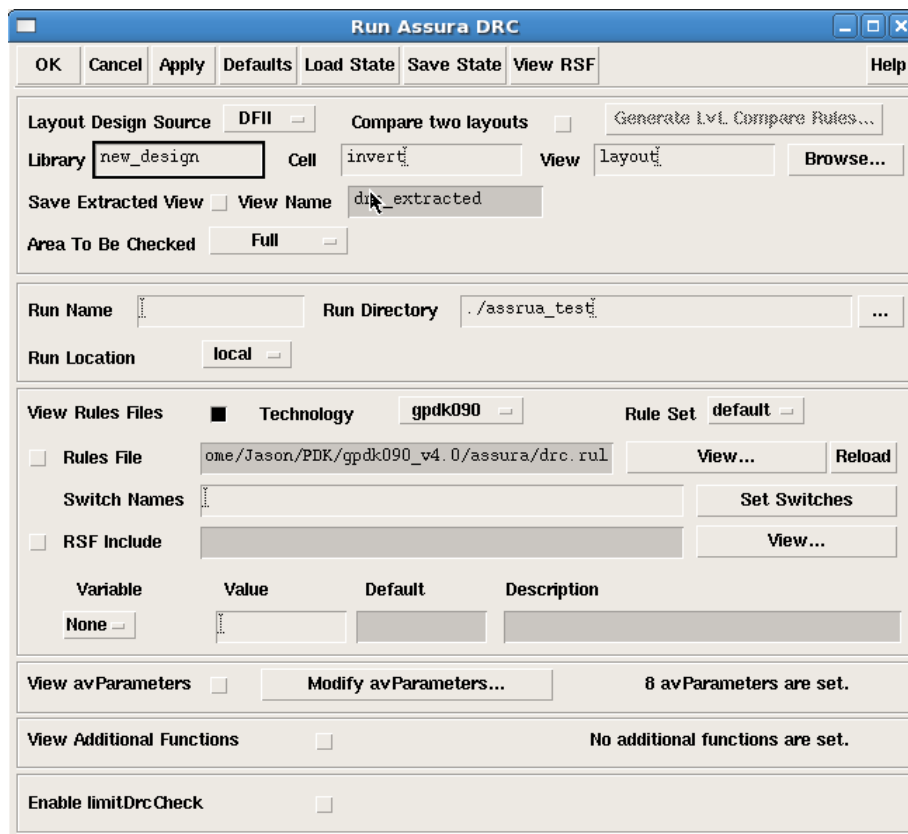


Fig.21

6.1.1.2 Assura DFII LVS

After the layout has no DRC violations (DRC free), the next step is to run the LVS check to make sure the layout is totally match to the schematic.

1. Click “Assura->Run LVS...” in layout window to invoke Assura LVS graphic user interface.
2. Fill in the “Assura run directory” and select the “Technology” field to “ssura_tech” (Fig. 22) in Assura LVS window. You can also set the Assura LVS switches and other parameters if needed.
3. Click “OK” to run the Assura LVS and see the result. If the layout isn’t matched to schematic, you have to manually re-edit the layout and re-run the LVS check to make the LVS result matched.

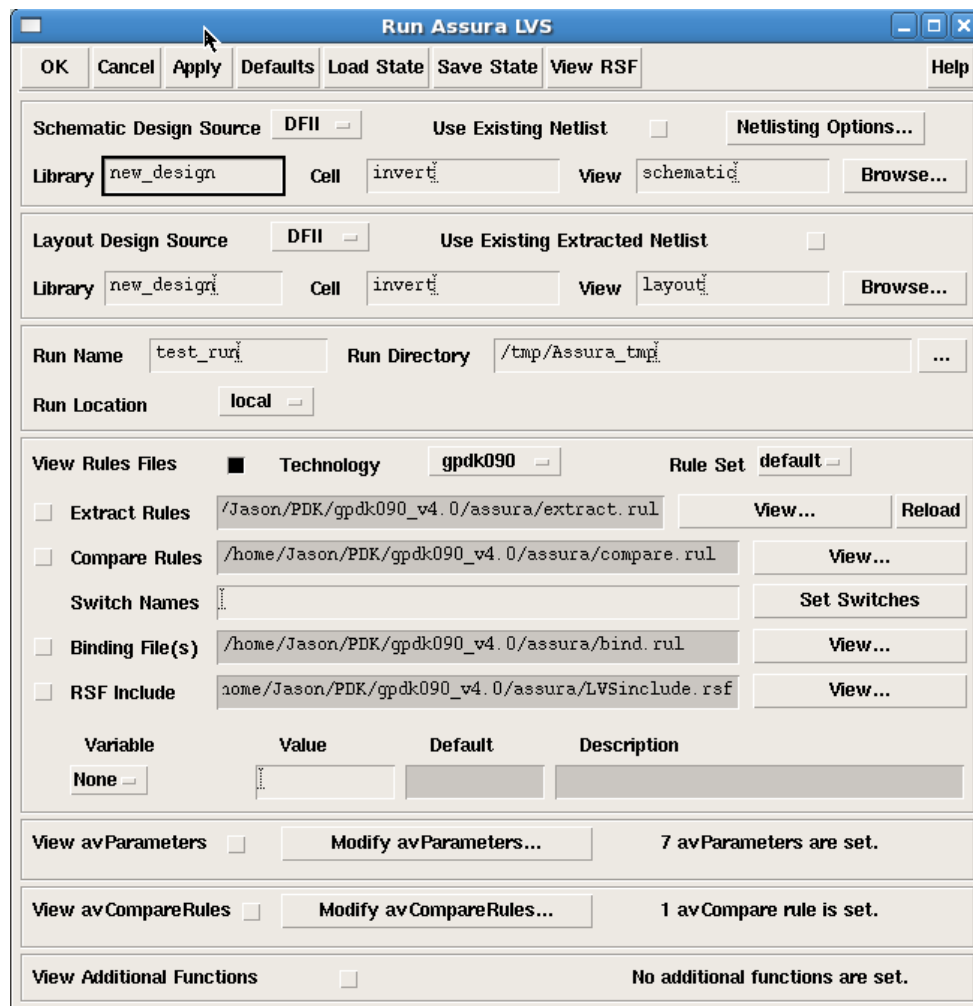


Fig.22

6.1.1.3 Assura DFII RCX

When the layout is DRC free and LVS clean, the next step is to perform the RC extraction. This step is to prepare the layout extracted netlist for post-layout simulation.

1. Click “Assura->Run RCX...” in layout window to invoke Assura RCX graphic user interface.
2. Select “Output” to “Extracted View” in “setup” folder of Assura RCX window to output extract result to “av_extracted” view (Fig 23).
3. In the “Extraction” folder of Assura RCX window, select the “extraction mode” to “C only” (if you want to extract only C), set the “Name space” to “Schematic Names” and fill in the “Ref Node”(here we use “gnd!”)(Fig 24).
4. Click “OK” to start the Assura RC extraction. After the RC extraction is completed, a new view (“av_extracted” view, Fig 25) which contains not only the original components but also the parasitic devices will be generated and then can be used for post-layout simulation.

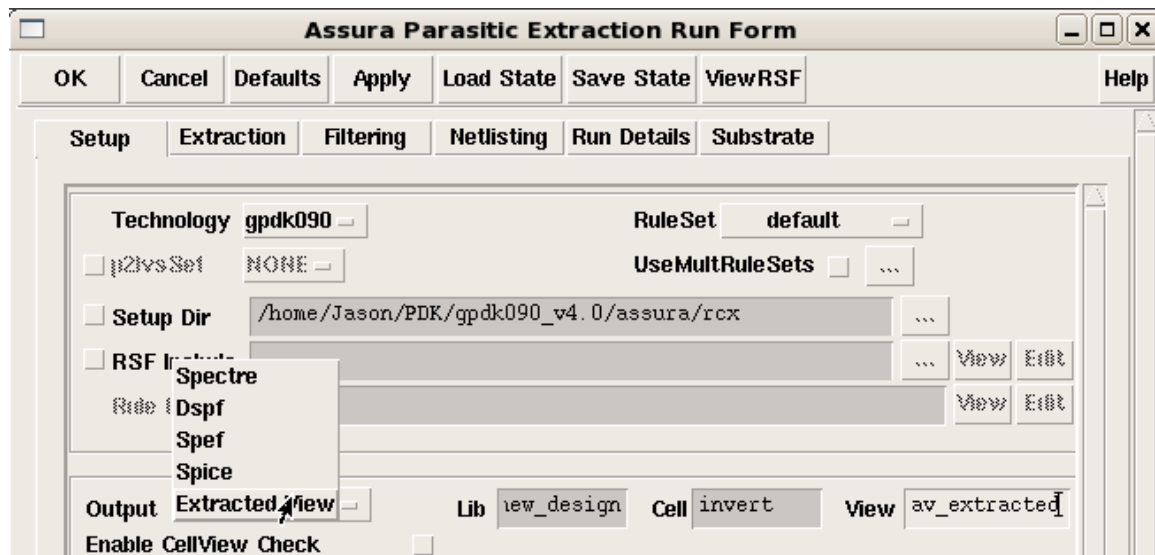


Fig.23

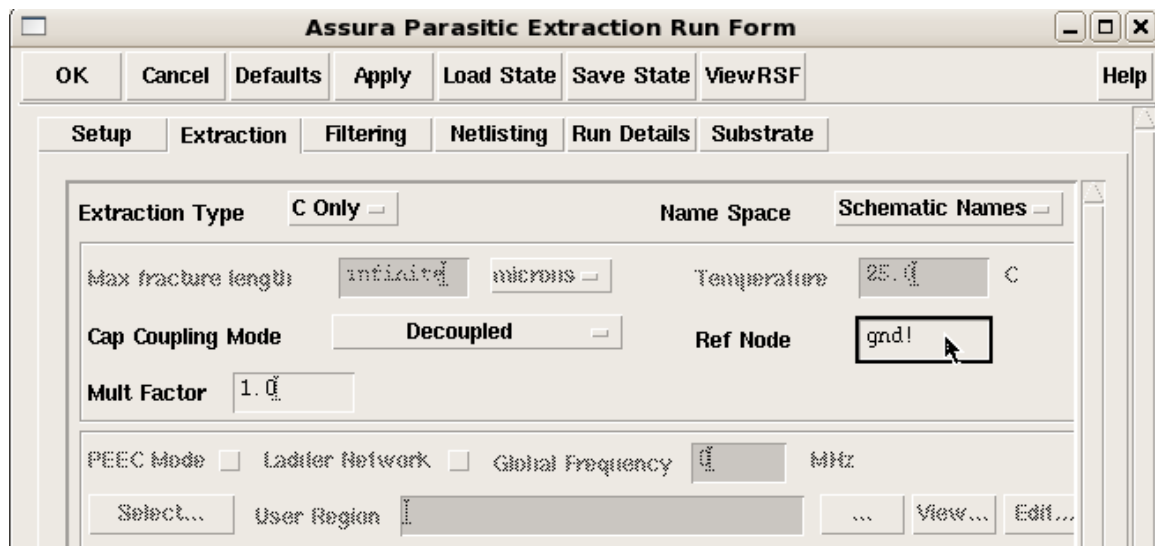


Fig.24

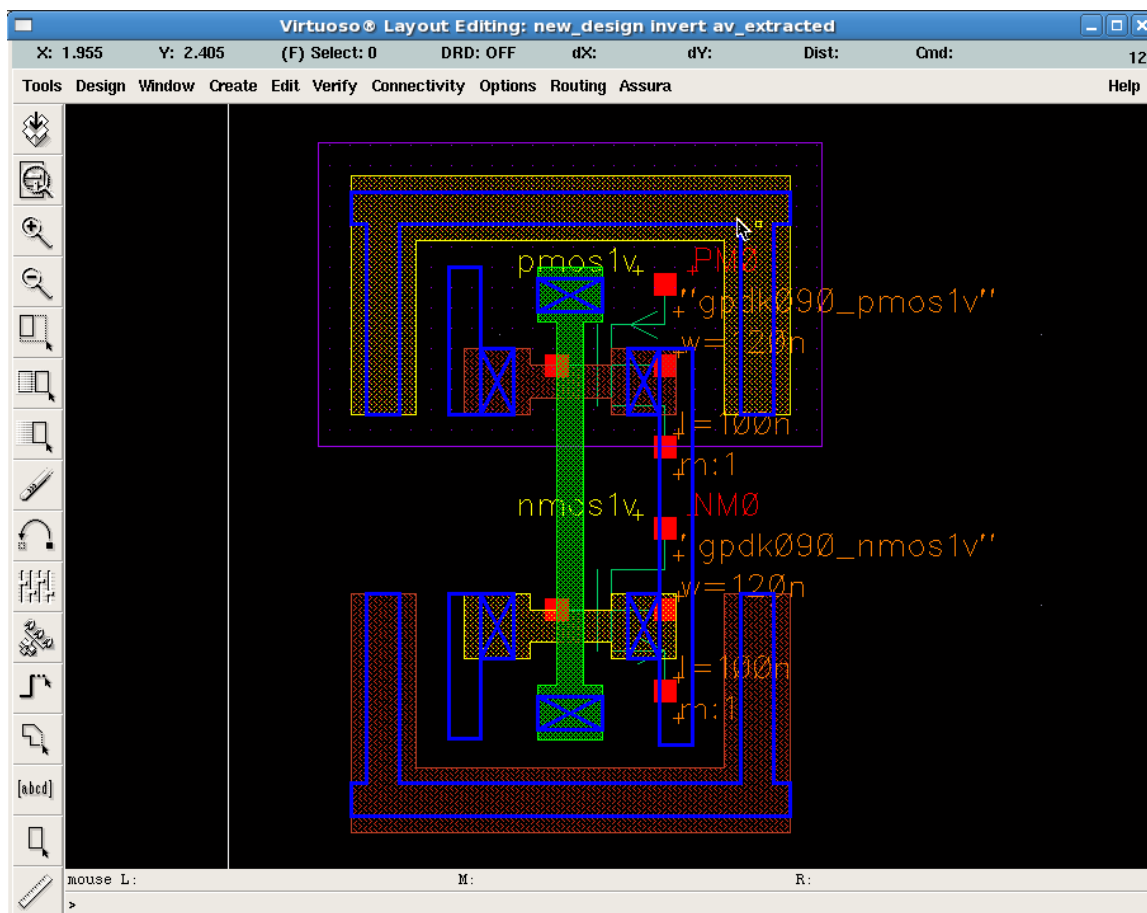


Fig.25

6.1.2 Assura CDL flow

In addition to the DFII flow, some users may use Assura CDL flow for their physical verification.

Currently, PDA's PDK only supports the Assura CDL flow in DRC and LVS verification.

6.1.2.1 Prepare Data

Before starting the Calibre CDL verification flow, users have to prepare the CDL netlist and the GDS file.

Export CDL netlist:

1. Select "File->Export->CDL..." from the CIW window to export the schematic netlist
2. Select the top cell name to your design, make sure the netlist mode is "Analog", and fill in the output file name of the netlist (Fig 26).

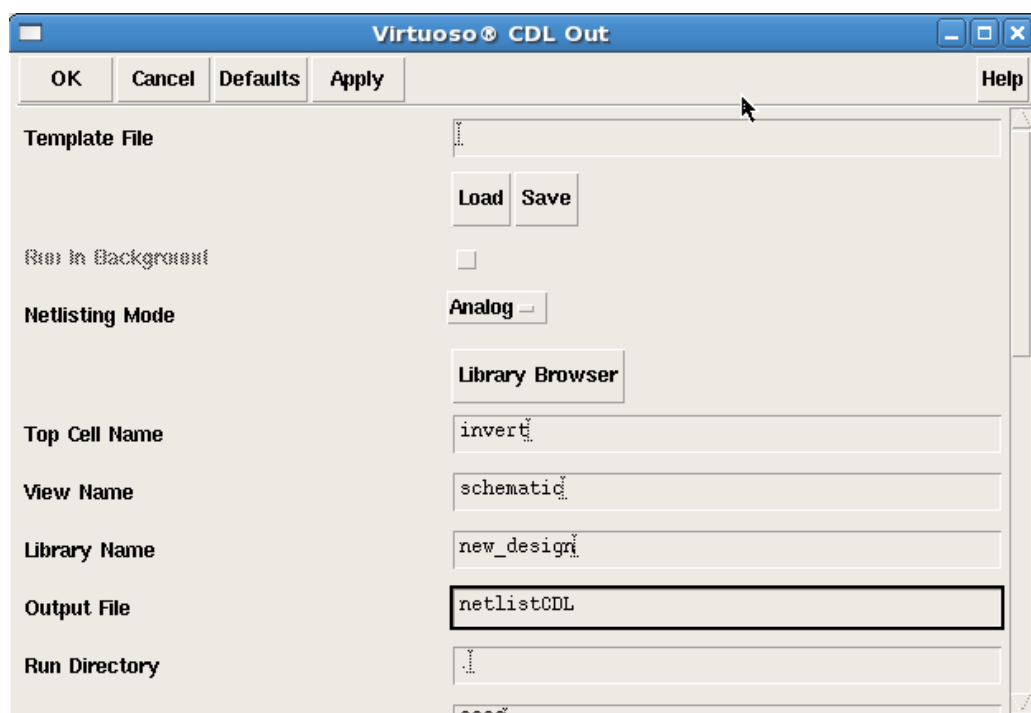


Fig.26

3. If the “source_added” file (empty sub-circuit file) is provided along with the LVS deck, you also have to attach this file to the output netlist file (obtained from step #2).

Export GDS file:

1. Select “File->Export->Stream...” from the CIW window to export the GDS file.
2. Select the top cell name to your design, fill in the “layer map table” and the output file name of the GDS file (Fig 27).

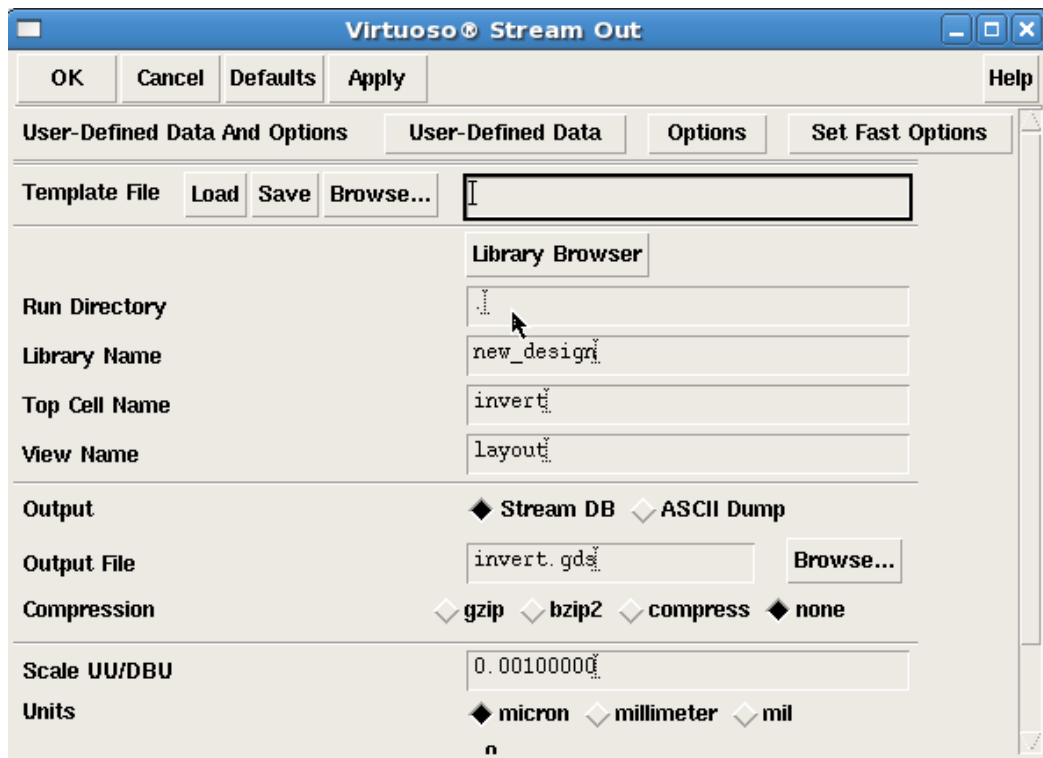


Fig.27

6.1.2.2 Assura CDL DRC

After the GDS file is exported, you have to edit the Assura ASCII run specific file (RSF) to specify the input data, output data and types of checks to be run in Assura DRC check.

1. Edit Assura DRC RSF file to specify the necessary information (Fig 28).

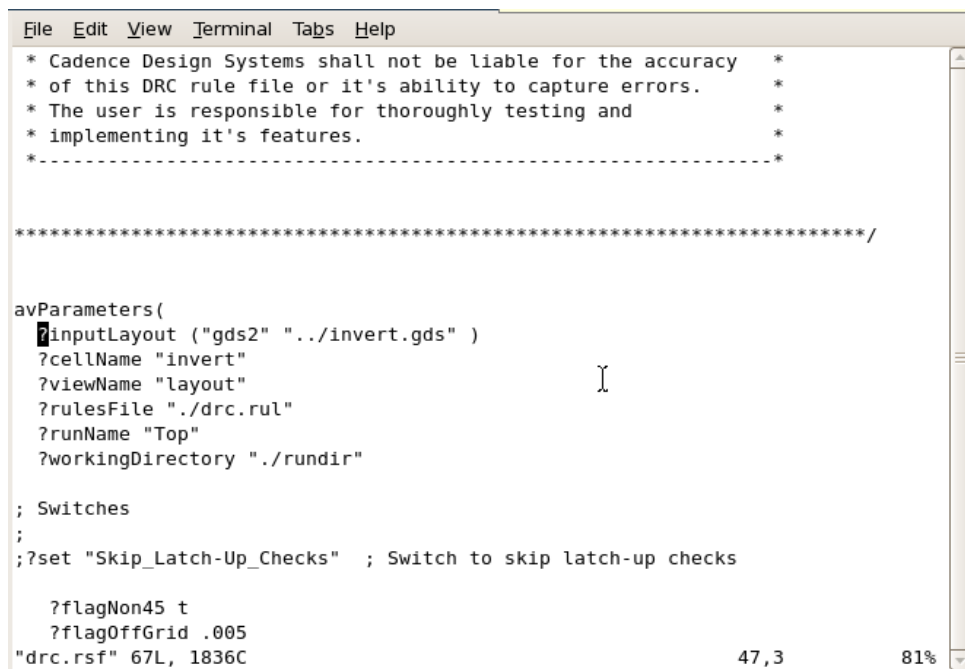


Fig.28

2. Run Assura DRC checking in UNIX command prompt and check the results.

%assura drc.rsf

3. If the GDS isn't DRC free, you have to go back to fix you layout and re-stream out the GDS for DRC check again.

6.1.2.3 Assura CDL LVS

When the GDS file is DRC free, you can continue to run the Assura CDL LVS.

1. Edit Assura LVS RSF file to specify the necessary information (Fig 29).
2. Run Assura LVS checking in UNIX command prompt and check the results.

%assura lvs.rsf

3. If the GDS isn't LVS clean, you have to go back to fix you layout and re-stream out the GDS for LVS check again.

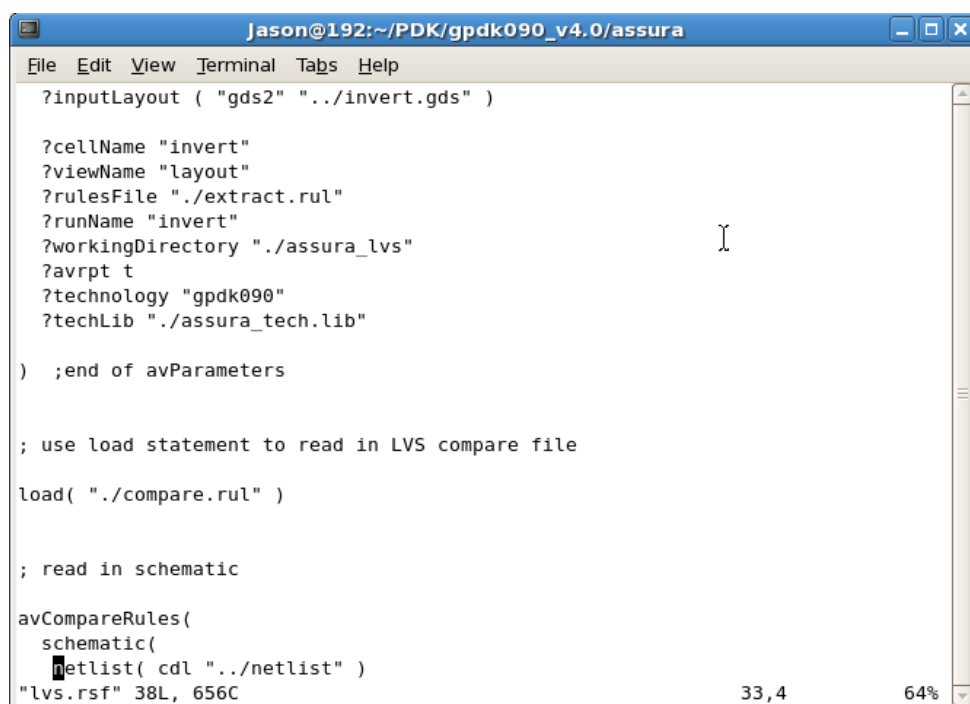


Fig. 29

6.1.2.4 Assura CDL RCX

Currently, PDA's PDK doesn't support the Assura RC extraction with CDL flow.

6.2 Physical verification using Calibre

PDA's PDK supports using Calibre for physical verification. In this section, we will introduce the "Calibre CDL flow" and "Calibre interactive flow" for Calibre users.

6.2.1 Calibre CDL flow

Before starting the Calibre CDL physical verification flow, users have to prepare the CDL netlist and the GDS

file first.

6.2.1.1 Calibre CDL DRC

After the GDS file is ready, you have to:

1. Edit the Calibre DRC deck and specify the GDS file name and top cell name for the deck .
2. Check the switches in the deck and correctly set these switches if need.
3. Run Calibre CDL flow DRC in UNIX command prompt and check the DRC results.

```
%calibre -drc -hier calibre.drc
```

4. If the layout isn't DRC free, you have to fix all DRC violations, re-stream out the GDS file and run the DRC check again.

6.2.1.2 Calibre CDL LVS

When the DRC checking is completed and the layout is DRC free, the next step is to perform the LVS checking.

1. Edit the Calibre LVS deck and specify the GDS file name, top cell name and CDL netlist file for the deck.
2. Check the switches in the deck and correctly set these switches if need.
3. Run Calibre CDL flow LVS in UNIX command prompt and check the LVS results.

```
%calibre -lvs -hier -spi layout.net calibre.drc
```

4. If the layout isn't LVS clean, you have to fix the layout, re-stream out the GDS file and run the LVS check again.

6.2.1.3 Calibre CDL RCX

After the layout is DRC free and LVS clean, the next step is to perform the RC extraction. This step is to prepare the layout extracted netlist for post-layoutsimulation.

1. Edit the Calibre RC deck and specify the GDS file name, top cell name and CDL netlist file for the deck
2. Check the switches in the deck and correctly set these switches if need.
3. Run Calibre CDL RCX in UNIX command prompt and obtain the extracted netlist. (Use extract C only for illustration)

```
%calibre -xrc -phdb calibre.drc
```

```
%calibre -xrc -pdb -c calibre.drc
```

```
%calibre -xrc -fmt -c calibre.drc
```

6.2.2 Calibre GUI flow (with Calibre-view)

If Calibre users don't like the CDL flow, PDA's PDK also support the Calibre GUI mode flow which integrate the Calibre interface into Cadence Virtuoso environment.

6.2.2.1 Calibre GUI DRC

Calibre GUI mode flow is working under Cadence Virtuoso environment and its DRC verification procedures are as follows:

1. Click "Calibre->Run DRC" in layout window to invoke Calibre DRC graphic user interface.

2. Specify the “Calibre-DRC rules file” and the “Primary cell” in Calibre interactive-DRC window (Fig 29). If you need to change some DRC switches, you have to edit the Calibre DRC deck first.
3. Click “Run DRC” button to start the Calibre DRC verification and check the result in RVE.
4. If the layout isn’t DRC free, you have to re-edit the layout and re-run the DRC check to make it DRC free.

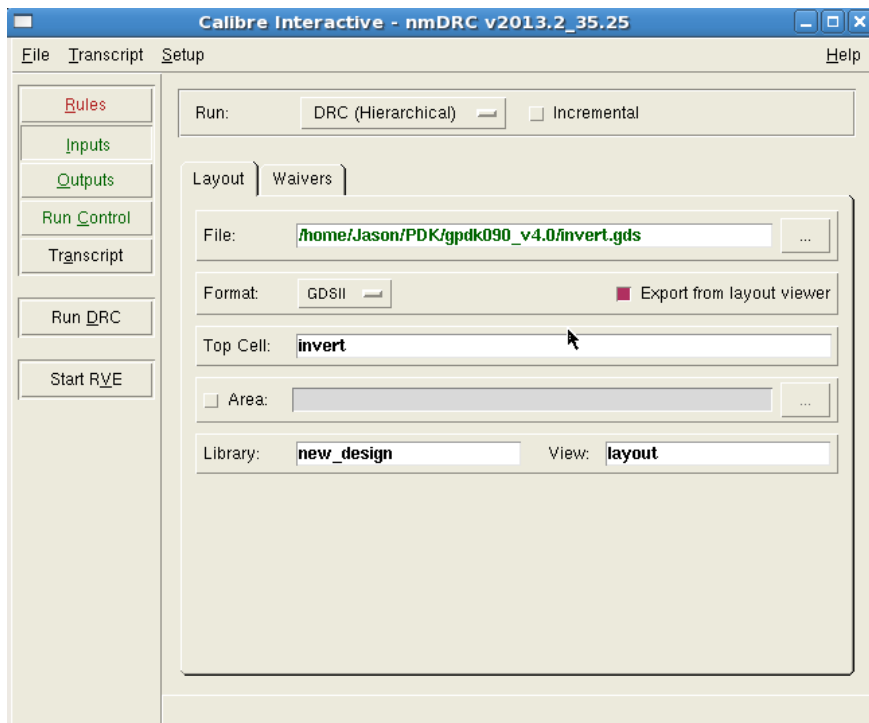


Fig. 29

6.2.2.2 Calibre GUI LVS

After the layout has no DRC violations, the next step is to run the LVS check to make sure the layout is completely match to the schematic.

1. If the “source_added” file (empty sub-circuit file) is provided along with the LVS deck, you have to specify this file as an include file for netlist export by click “Calibre->Setup->Netlist Export...” (fig 30).
2. Click “Calibre->Run LVS” in layout window to invoke Calibre LVS graphic user interface.
3. Specify the “Calibre-LVS rules file”, working directory and “Primary cell” in Calibre LVS window. If you need to change some LVS switches, you have to edit the Calibre LVS deck first.
4. Click “OK” to run the Calibre GUI LVS and see the result. If the layout isn’t matched to schematic, you have to fix the layout and re-run the LVS check to make the LVS result matched.

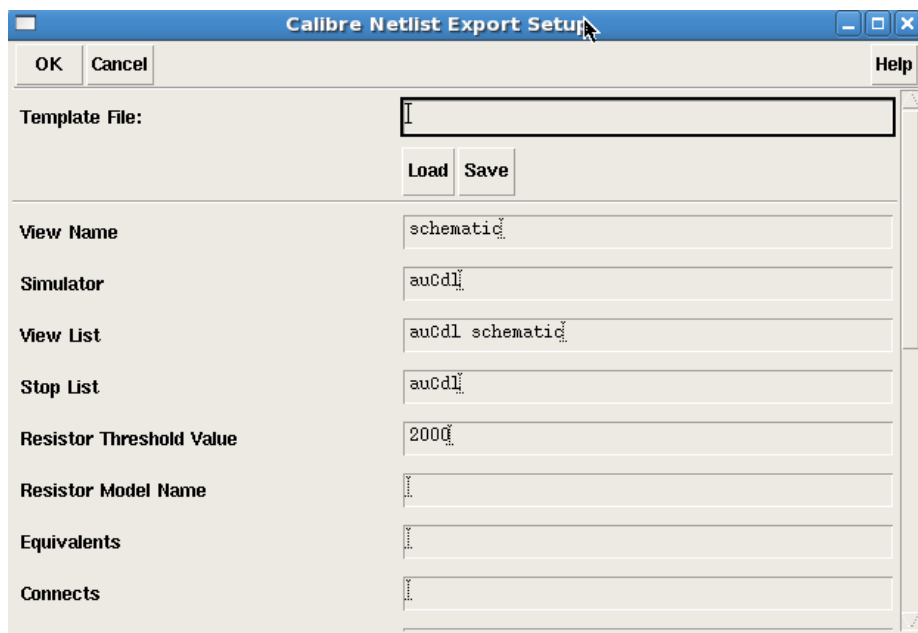


fig. 30

6.2.2.3 Calibre GUI RCX (with Calibre-view)

When the layout is verified to be DRC free and LVS clean, the next step is to perform the RC extraction. In addition to directly obtain the ascii extracted netlist file from Calibre GUI mode extraction (like Calibre CDL RCX), you can also choose to obtain an extracted view 'calibre-view' which is somewhat like the Assura av_extracted view. In this calibre-view, you can see not only the original components and but also the parasitic devices and their connectivity. Furthermore, this calibre-view can also be used in Cadence Analog Artist environment for post-layout simulation.

1. Click "Calibre->Run PEX" in layout window to invoke Calibre RCX graphic user interface.
2. Specify the "Calibre-PEX rules file", working directory and "Primary cell" in Calibre RCX window. If you need to change some RCX switches, you have to edit the Calibre RCX deck first.
3. Select the "Extraction Type", specify the output format to be "CALIBREVIEW" and "Use Names From" to be "SOURCE" (fig 31).
4. Click "OK" to run the Calibre GUI RC extraction. After the extraction run is completed, a calibre view setup window pops up. Specify the "Cellmap File", "Magnify symbols by" and make sure the "Copy schematic properties" button is on.
5. Click "OK" in the Calibre view setup window to create the Calibre view.

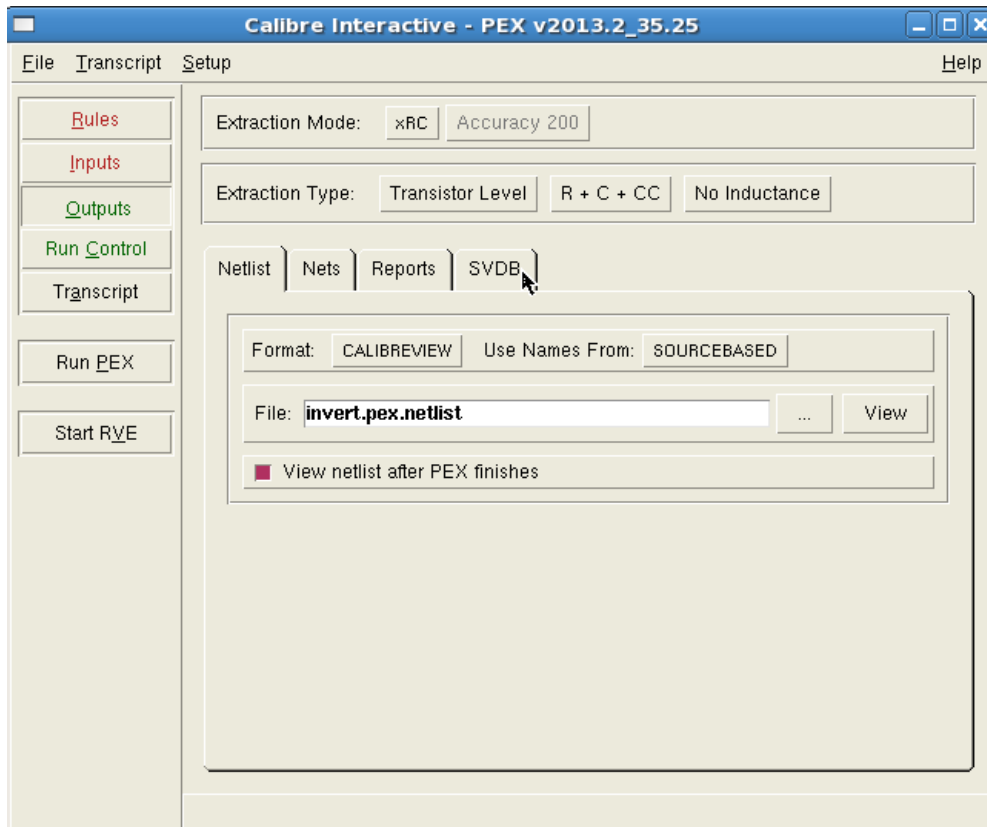


Fig.31

7 Parasitic re-simulation

When accomplished the physical verification, the last step to tape-out is to perform the post-layout simulation on the extracted netlist/view. During the post-layout simulation, not only the original components but also the parasitic R & C (depends on what you have extracted in RCX stage) of the interconnections are taking into consideration. Therefore, we can say that the post-layout simulation result is more approach to the real silicon comparing with the original pre-layout simulation result. Furthermore, based on the difference of RC extraction flows you chose you would run your post-layout simulation in different ways.

7.1 Simulating with the extracted view

If you choose to output an extracted view during the RC extraction phase (such as go through “Calibre GUI RCX” flow or “Assura DFII RCX” flow), you may plan to run the post-layout simulation with GUI mode in Cadence Analog Artist environment. Here, we use the “av_extracted” view, which is obtained from “Assura DFII RCX” as an example, and run the post-layout simulation with Spectre simulator in Cadence analog Artist environment.

1.Create a config view for the test fixture schematic (the ‘test_invert’ in our example) and switch the av_extracted (or calibre) view of our design (invert) for the configuration (Fig 32).

Note: The detail on creating a config view in Cadence Virtuoso environment should be refered to Cadence Analog Design Environment manual.

2.Open the Analog Artist environment and use the config view of the test fixture for simulation (fig 33). You can also output the simulation netlist to make sure the ‘av_extracted’ view of our design in the configuration file is used for simulation rather than the ‘schematic’ view.

3.Run the post-layout simulation and check the simulation result.

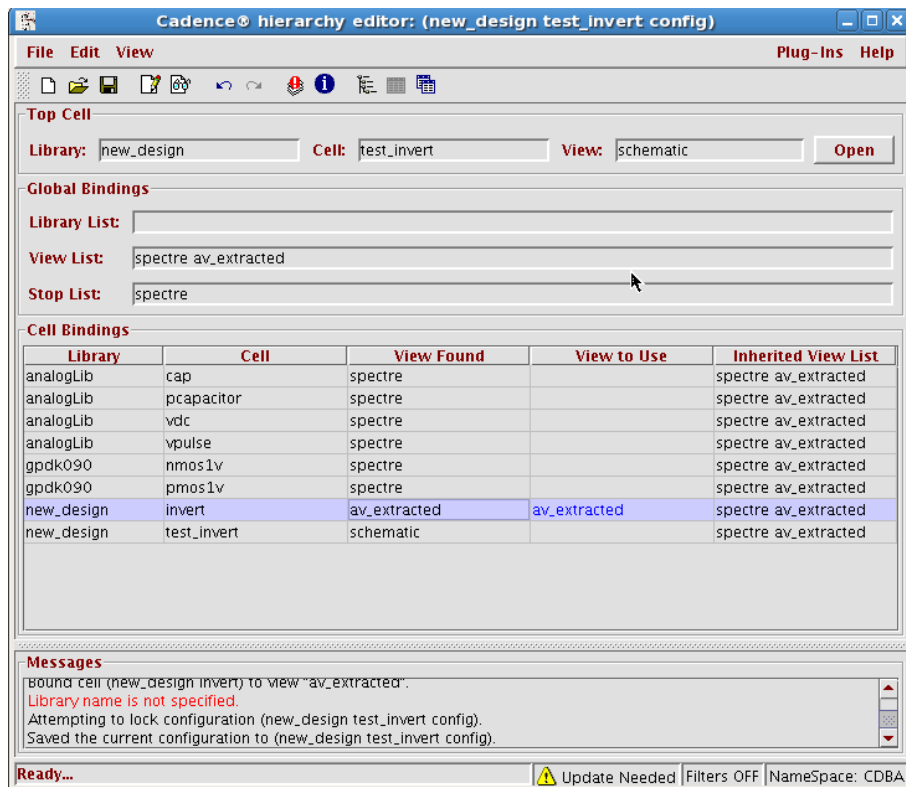


Fig 32

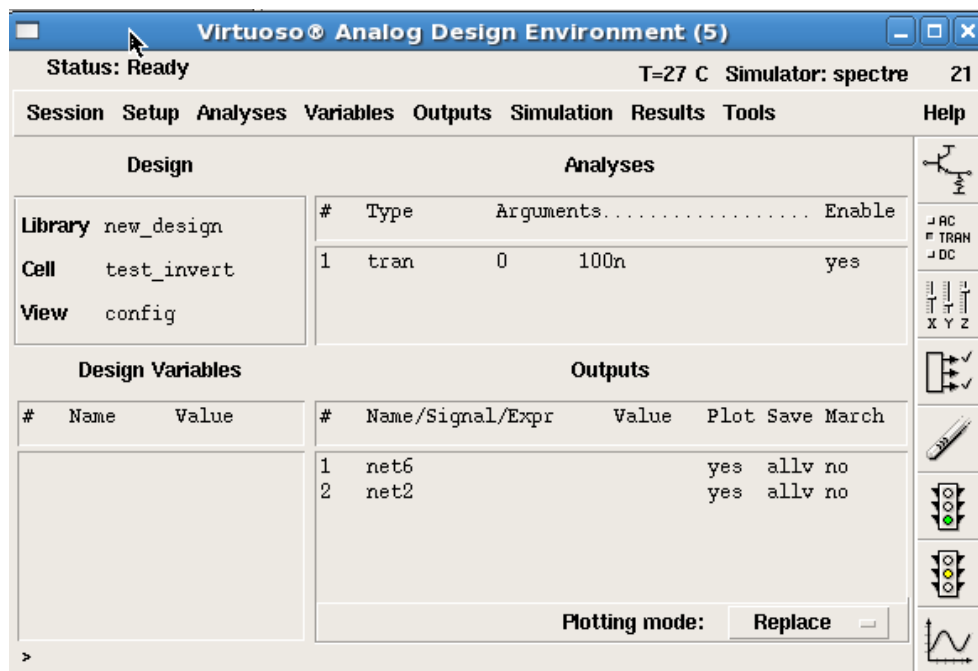


Fig.33

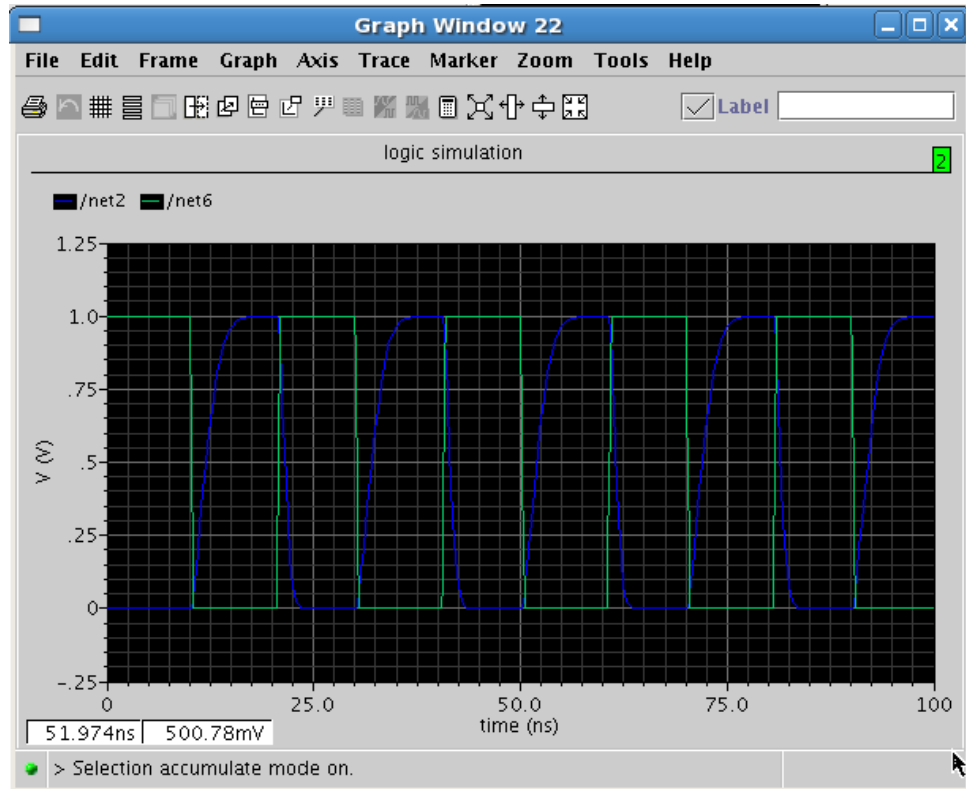


Fig.44